

Early-Warning Signals of Regime Shifts in Euro-Area Sovereign Spreads

A Complexity-Theory Approach

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Glossary of Key Terminology

System	<i>the sovereign bond market treated as a whole: the network of interacting investors, institutions, and prices whose collective behaviour generates spreads</i>
State	<i>the condition the system is in at any given time, defined by its statistical properties: calm or stress, stable or fragile</i>
Regime Shift / Critical Transition	<i>a large, often irreversible change in how a system behaves: not just a big move in prices, but a structural break where the rules governing the system change</i>
Tipping Point	<i>the threshold at which a small additional shock causes the system to transition into a new regime. Once crossed, the shift is typically self-reinforcing and difficult to reverse</i>
Unstable Equilibrium	<i>a state the system passes through during a transition where any small perturbation sends it sharply in one direction or the other</i>
Endogenous Fragility	<i>vulnerability that builds gradually from within the system itself, through the interactions of its own participants, rather than an external shock</i>
Contagion	<i>the spread of stress from one part of a financial system to another through linkages (sovereign to bank, country to country) amplifying shocks beyond their point of origin</i>

What is Complexity Theory and Why is it Important?

Traditional Equilibrium Models

LIMITATIONS

- **Markets revert to stable equilibrium**
Assumes any shock is absorbed and prices return to a long-run steady state; ignoring how crises can permanently alter market structure
- **Cannot model feedback loops or contagion**
Linear frameworks miss self-reinforcing dynamics where distress in one area amplifies and propagates through interconnected networks
- **Crises explained by exogenous shocks**
External events (wars, pandemics) are invoked as causes, ignoring the endogenous fragility that builds gradually inside the system itself



Complexity Theory Approach

WHAT IT OFFERS

- **Economy as a dynamic, bottom-up system**
Heterogeneous agents continuously adapt their beliefs and strategies, generating emergent macro phenomena without any central coordinator
- **Crises emerge from endogenous fragility**
Shocks spread through financial networks via contagion and self-reinforcing cycles; a gradual reduction in the system's ability to absorb shocks means that eventually, even a minor disturbance can trigger a regime shift
- **Critical slowing down as early warning**
Before a tipping point, systems recover slower from disturbances. Rising autocorrelation, variance, and skewness are detectable signals of deteriorating resilience

***Application to sovereign bond spreads:** they are shaped by interacting forces and often exhibit abrupt, endogenous shifts with stress building invisibly before it materialises. This makes them well suited to complexity-based analysis and early-warning signal detection.*

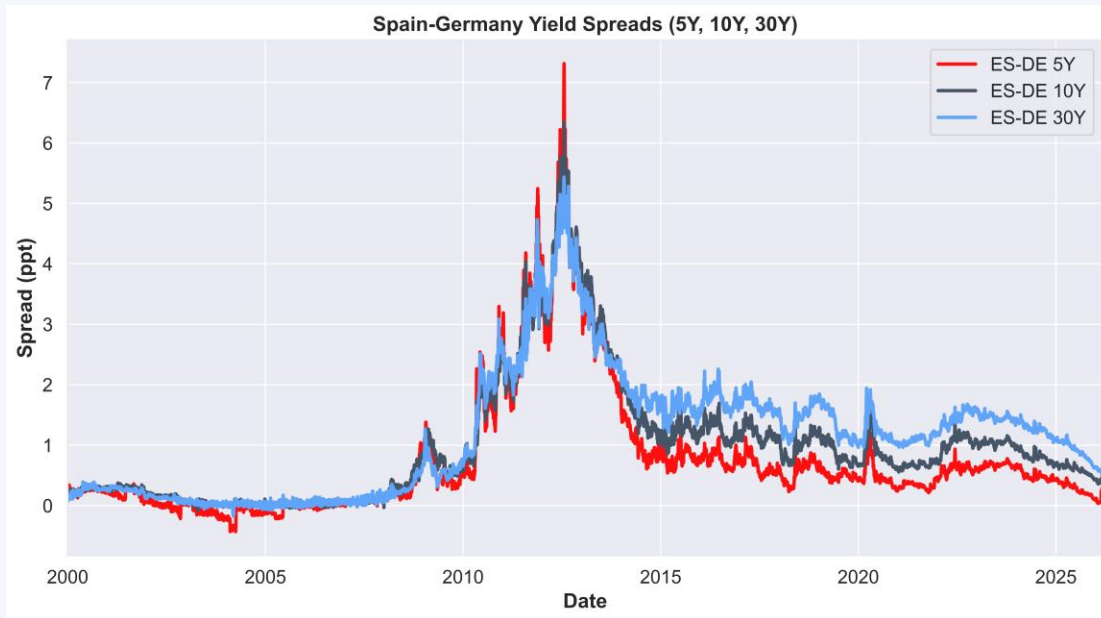
Data: Why Spain-Germany 5-Year Spreads?

Ideal Test Case: Spain

- 4th-largest euro-area economy: large enough for systemic implications, yet no full sovereign bailout.
- Complete crisis arc: sharp deterioration →

Sensitivity to Risk: 5-year tenor

- Liquid, actively traded point on the yield curve.
- Provides the cleanest signal of sovereign credit conditions, minimising idiosyncratic pricing noise.



German Bunds are the accepted safe-haven reference point in European fixed income markets. As per standard practice, we consider the Spain-Germany spread to directly measure the market's perception of Spanish sovereign risk.

Five Complexity-Based Indicators

Indicator	Mathematical Definition	Economic Explanation	Warning Signal When...
Critical Slowing Down	$\tau = -\ln(\rho)$	Decay rate of small perturbations; system recovers slower from disturbances as tipping point nears	Fall in τ towards 0 (lower 10% tail)
Autocorrelation	$N_t = c + \rho N_{t-1} + \varepsilon_t$	Rise in serial persistence near critical thresholds; current state increasingly resembles recent past	Rise in ρ (upper 10% tail)
Variance	$\sigma^2 = \frac{1}{n-1} \sum (N_t - \mu)^2$	Accumulated impact of shocks; fluctuations amplify because slower decay rates allow variances to build	Rise in σ^2 (upper 10% tail)
Skewness	$\gamma = \frac{m_3}{\sigma^3}$	Distribution becomes asymmetric as system nears unstable equilibrium	Rise in γ (upper 10% tail)
Flickering	$\beta = \frac{\gamma^2 + 1}{\kappa}$	Bimodal distribution; system alternates between old and new state	Rise in β towards 1 (upper 10% tail)

From Indicators to an EWS Score



Methodology Explanation

- **100-day window:** balances sufficient data for stable estimation against responsiveness to recent dynamics
- **5 indicators:** no single measure is reliable alone; co-movement across multiple reduces false positives
- **10% tail:** fixed modelling assumption. Thresholds are retrospective, not real-time
- **EWS score:** a single composite count makes cross-episode comparison tractable
- **≥3 threshold:** requiring a majority to fire simultaneously guards against noise-driven false flags

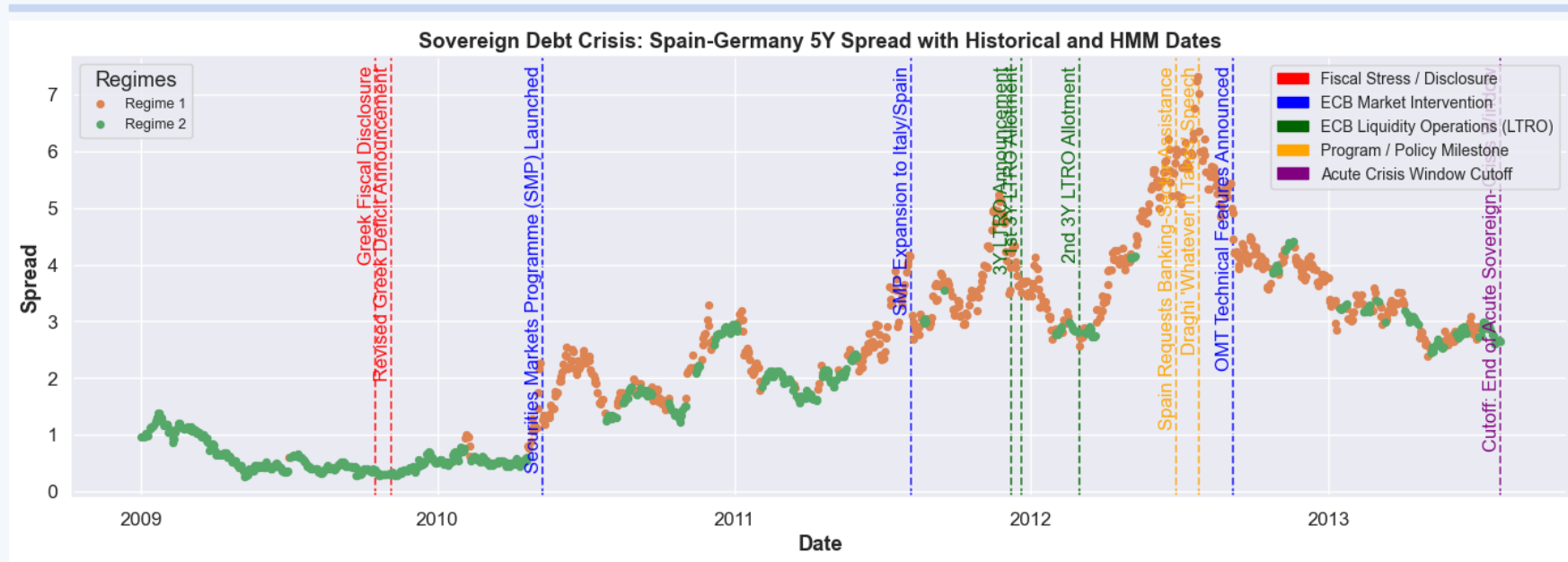
Note that these modelling assumptions are consistent with van den End (2019)

Five Episodes, Not Individual Events

Nearby events driven by a common shock are consolidated around a single anchor date to avoid artificial clustering.

01	Greek Fiscal Shock 16 Oct 2009	Exogenous	Greece reveals deficit at roughly double what had previously been reported, undermining the credibility of EU fiscal surveillance. This fundamentally alters the pricing of peripheral sovereign risk across the eurozone, serving as an external shock to Spanish spreads.
02	First Eurozone Intervention (SMP) 10 May 2010	Endogenous	ECB launches its first direct sovereign bond purchase programme, buying government debt from distressed countries to restore liquidity to dysfunctional markets and address vulnerabilities to self-fulfilling liquidity crises.
03	Spain Contagion 7 Aug 2011	Endogenous	ECB explicitly extends its bond purchases to Spain for the first time as Spanish spreads reach unsustainable levels. Following sequential bailout requests from Portugal and Greece, it is clear that contagion has spread well beyond Greece.
04	LTRO Liquidity Episode 8 Dec 2011	Exogenous	ECB announces 3-year LTROs, offering eurozone banks three-year loans at fixed rates, ultimately injecting around €1 trillion into the banking system. This is a discretionary policy resulting from institutional intervention, not the product of endogenous fragility.
05	Spanish Banking Assistance 27 Jun 2012	Endogenous	Spain requests EU financial assistance for its banking sector, crystallising the doom loop as bank bailout costs transfer directly onto sovereign balance sheets. A subsequent surge in Spanish yields was only halted by Draghi's "whatever it takes" speech.

Statistical Cross-Check: Calm vs Stress Regimes



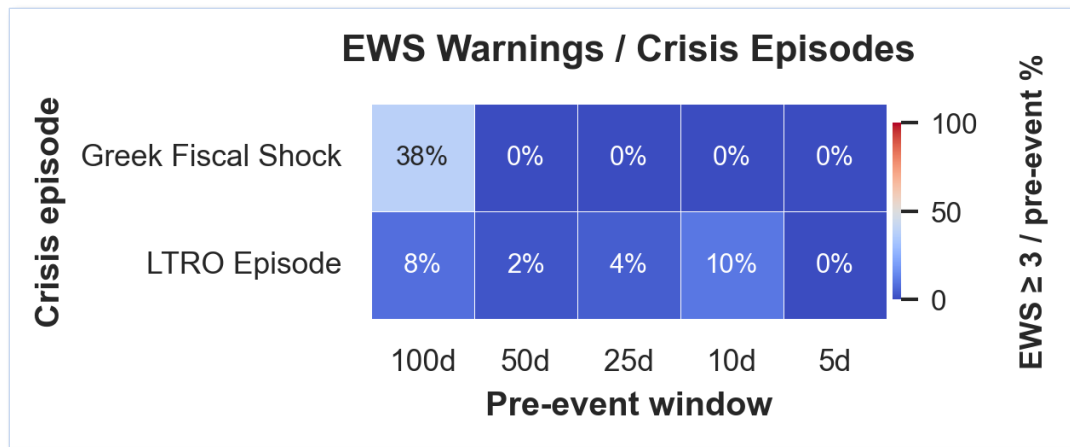
METHOD

- **Regime 1 = Stress; Regime 2 = Calm**
- Two-state Markov-switching **AR(1)** model
- States inferred from the 5Y spread's **own dynamics**

OUTCOME

- Mean spread: **3.39 vs 1.40**
- **Persistent** Regimes, **≈94% probability** of staying in same state
- Stress states overlap mainly with **2011–2012 crisis escalation**

Exogenous Events: Greek Fiscal Shock & LTRO Episode



Greek Fiscal Shock: 16 Oct 2009

- Virtually no meaningful EWS signalling: zero EWS scores over 3 in the 50 days immediately before the anchor date
- Spreads narrowed to ~ 0.4 as Greece disclosed: no pre-transitional state

LTRO Liquidity Episode: 8 Dec 2011

- Very limited EWS activity: sporadic signals and near-total absence in final period, EWS score = 0 on last observation
- Discretionary policy intervention, not the product of endogenous spread fragility approaching a tipping point

Exogenous Events: Greek Fiscal Shock & LTRO Episode



Interpretation

- Neither episode constitutes a methodology failure, both are informative contrasts; the absence of EWS activity correctly reflects that Spanish spreads were not approaching a critical transition in either case
- The Greek disclosure was an exogenous shock to Spanish spreads; the LTRO a discretionary policy decision
- The framework discriminates accurately as there was no culmination of endogenous fragility → silence is the correct prediction

Endogenous Events: SMP Launch, Spain Contagion, Banking Assistance

SMP Launch: 10 May 2010

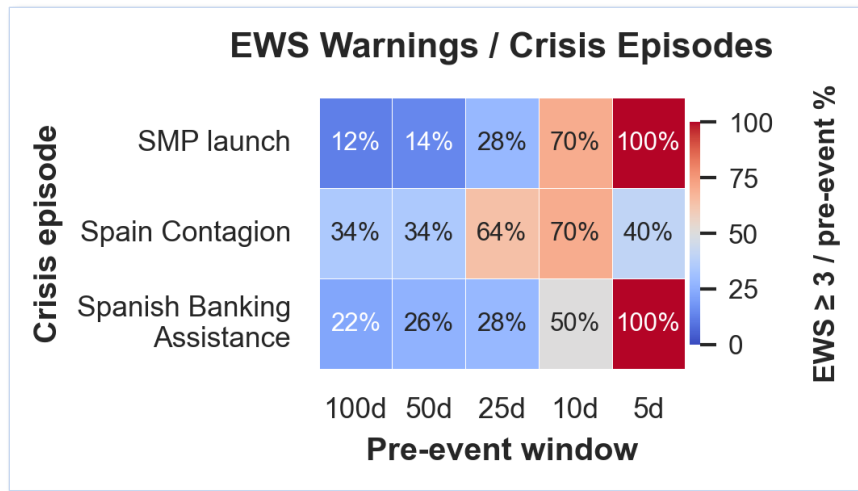
- Strong and concentrated EWS activity: sparse signalling until marked intensification in the final days
- Sharp jump from EWS score of 1 just days before
- Signals rapidly clustering and intensifying consistent with a tipping point

Spain Contagion (SMP Expansion): 7 Aug 2011

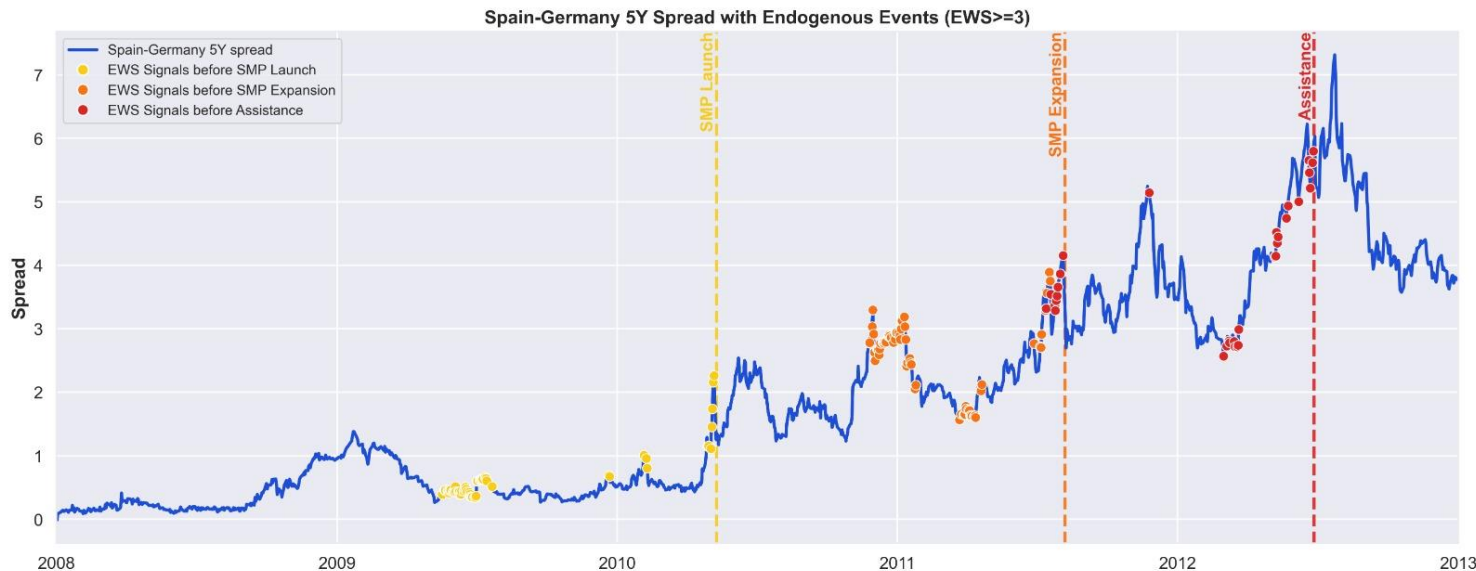
- Sustained and multi-horizon EWS activity: frequent scores above 3 across all time windows
- EWS oscillated on rising spread trajectory, consistent with gradual endogenous build-up to a critical transition

Spanish Banking Assistance: 27 Jun 2012

- Increasingly consistent EWS activity leading up to event: regular flags as spreads ticked upwards
- EWS activity began at the low point of spreads in March 2012 following the LTRO calm period



Endogenous Events: SMP Launch, Spain Contagion, Banking Assistance



Interpretation

- All three endogenous episodes were preceded by meaningful EWS activity
- Endogenous fragility builds measurably, so the framework is best suited to detect self-reinforcing spread dynamics where instability accumulates ahead of a regime shift
- The distinction between endogenous fragility and exogenous shocks is central to determining what the EWS framework can and cannot be expected to detect

EWS Signals Mark Fragility, Not Spread Direction

Horizon (h)	Signal μ	Random μ	Signal σ	Random σ	P-value	Interpretation
5 days	-0.01	0.02	0.36	0.11	0.439	No
10 days	-0.08	0.03	0.43	0.16	0.008	Yes (5% level)
15 days	-0.08	0.04	0.60	0.19	0.029	Yes (5% level)
20 days	-0.07	0.04	0.69	0.24	0.076	Yes (10% level)
25 days	-0.10	0.03	0.72	0.24	0.054	Yes (10% level)

Welch's Two-sided T-test:

$$H_0: \mu_{\text{signal},h} = \mu_{\text{random},h}$$

$$H_1: \mu_{\text{signal},h} \neq \mu_{\text{random},h}$$

- **Negative** Mean Differences
- Spreads **stabilise/narrow** after signals, not widen further

- **EWS** = Fragility Marker, Not Direction
- Signals identify periods of instability
→ Institutional intervention

Note: μ and σ are measured in percentage points; 0.08 pp = 8 bp.

Limitations and Next Steps

Limitations

- **10% tail threshold:** modelling assumption, not estimated from data
- **Ex-post analysis:** thresholds are retrospective, not real-time warnings
- **HMM provides a useful statistical benchmark,** although its parameter estimates remain sensitive to model specification and sample selection.
- **Dependence of observations:** clustered signal dates and overlapping post-signal windows reduce independence of observations

Future Research

- **Apply to Portugal and Italy:** isolate common systemic dynamics from Spanish idiosyncrasies
- **Extending the analysis:** emerging market sovereign spreads and exchange rate regimes (EURCHF), where tipping-point dynamics can be more pronounced
- **Examine policy reaction:** do policymakers respond to the same signals detected by EWS?
- **Combine EWS with other stress indicators:** test whether complexity signals provide additional insights when considered alongside other indicators.

Key Finding

"Complexity-theory EWS indicators have the potential to detect the build-up of endogenous fragility in sovereign spread markets ahead of regime shifts but are informatively silent when shocks are exogenous."

Thank You

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Questions?